外太空探索

是否存在外星智力?

LIFE ON OTHER PLANETS

1 For many years, people have **wondered** whether we are the only living things in the universe. Some scientists believe that there must be life on other planets because the universe is so big. However, it is **unlikely** that there is life on other planets because planets need a very specific environment for life to start. In the end, there are no facts that **support** the idea of life on other planets.

2 It is true that the universe has thousands of solar systems and there are planets similar to Earth. During its nine years in space, the very powerful Kepler telescope[1] found more than 2,600 planets in orbit around stars. In fact, a number of scientists think that one of these planets, named Kepler 22b, has the right **conditions** – the right atmosphere[2] and temperature – to support life. However, there is no **evidence** that there is life on Kepler 22b. Experts argue that even with the best technology we can see no signs of life there. Until research shows us hard evidence, we cannot use Kepler 22b to support the idea of life on other planets.

3 A planet needs very **particular** conditions to support life. A planet with life would need to have water, the right temperature and the right mix of chemicals in the atmosphere. Earth has the perfect conditions for life, and it is very unlikely that another planet has exactly the same environment as Earth. In addition, although some experts argue that life might **exist** on other planets, they have never found evidence to **prove** it. A report from Princeton University suggests that it is highly unlikely that there is life on other planets. The researchers believe that we don't have enough scientific evidence to decide if there is life on other planets. They say that just because conditions similar to Earth exist on other planets, it doesn't mean that life exists there.

4 In conclusion, I do not believe that there is life on other planets. Although the universe is very big, a planet with life needs very special conditions. Earth has exactly the right conditions for life. It is not too hot or too cold. It has water, air and the right chemicals. I do not think that any other planets could have exactly the same conditions as Earth. Therefore, I do not think that there could be life on other planets.

[1]**telescope** (n) a piece of equipment, in the shape of a tube, that makes things that are far away look bigger or nearer
[2]**atmosphere** (n) the layer of gases around a planet

word lists

考核频率	中文	英文
	有力的证据	hard evidence
	不太可能	it is unlikely that

main idea: 没有证据证明有外星生命
要不要花钱在外太空探索上面
没有证据证明有外星生命 - 探索： 徒劳

探索火星的可能性
Scientists are considering the possibility of sending humans to Mars in the coming decades. Although there have been successful manned missions to the Moon in the 1960s and 1970s, Mars is 150 times further away from Earth than the Moon is. Thus the project of sending humans to Mars would require solving an array of problems the Moon missions did not have to face.

One problem is that a round-trip to Mars and back is likely to take at least two years. The trip to the Moon lasted only a few days, and it was easy to bring enough supplies of food, water, and oxygen; but the cargo capacity of space vehicles is too limited to put on board the food, water, and oxygen required by a crew for a period of two years. Without those essentials, though, a Mars mission is impossible.

A second problem is that astronauts on the Mars mission would be in the zero-gravity environment of space for many months at a time. Spending a long time in the zero-gravity environment has negative effects on the human body, such as decreased muscle mass and lower bone density. Over the course of a two-year mission, the effects would be so severe, they would make it impossible for humans to make the trip without experiencing grave medical problems.

Finally, astronauts on a mission to Mars would be exposed to dangerous levels of space radiation, much of which comes in the form of charged particles emitted by the Sun. Earth's magnetic field, which normally protects us from dangerous solar radiation, would not be able to protect a spaceship traveling in interplanetary space. Constructing a shield that would protect the whole spaceship from space radiation is at present impossible because it would add too much weight to the ship.

驳斥段的思路
当有问题， 你提出解决措施
problem - solution

反方驳斥

A trip to mars would definitely be challenging but scientists have proposed solutions to the problems the reading selection discusses.

First of all, food, water and oxygen. Well astronauts can use hydroponics. Hydroponics is a technique for growing plants with their roots in water rather than in soil. It requires relatively little space. Using hydroponics, the astronauts should be able to cultivate food crops in the spacecraft. In addition, the hydroponic will grow plants with recycle waste water and release it as clean water vapor, which can be collected as drinking water. And of course, all plants absorb carbon dioxide and release oxygen. So thanks to hydroponics, the astronauts will also have fresh air to breathe.

Second, the effects of zero gravity. Over the last few decades, we have launched several space stations orbiting the earth. And a number of astronauts have spent many months on them in a zero gravity environment. These astronauts have learned to use several techniques to safely manage the effects of zero gravity. For example, regular exercise prevents the decrease in muscle mass. Likewise, taking vitamins and minerals like calcium slows down the decrease in astronauts' bone density.

Third, solar radiation. Astronauts traveling to Mars will be exposed to some solar radiation but this radiation will not be at dangerous levels all the time. The sun only releases dangerous amounts of radiation occasionally, during periods when it is particularly active. In order to avoid this threat, the spacecraft could be equipped with special instruments that monitor solar radiation and with a small shelter that shield against radiation but doesn't add much weight to the ship. Most of the time, the astronauts will go about their normal business in unshielded areas of the spacecraft. But when their instruments detect increased radiation, they could stay in the small shielded area until the danger has passed.

passage 3

C

Source: A government agency

Uncovering the mysteries of space is a huge task that should continue to be funded. Imagine the discoveries and advances in technology that will be made as countries go farther and farther into space, especially in the race to be the first to Mars. Furthermore, people may need to live on Mars someday because Earth will become too crowded and too hot. Also, asking questions about the universe encourages young people to study Science and Engineering, which is a huge benefit to society. Medical advances are also made as a result of sending humans to space. For example, when astronauts are in space, their bones become weaker and more likely to break. Many elderly people also have weak bones. Drugs that can help elderly people can be tested on astronauts in space. Research done in space can improve life on Earth. Governments should absolutely spend more money on space programmes.

TOPIC SENTENCE

科技进度 advances in technology
地球不适合居住 migrate to Mars -- too crowded \ too hot
鼓励年轻人学习工程 study science and engineering
医学进步 medical breakthrough - bone density

support
科学的突破 countries go further and further into space -- especially in the race to be the first to Mars --- make discoveries and advances in technology --- 举例？ --后面有一篇文章直接在讲解到底有哪些技术上的突破 --

医疗上突破对地球人有好处 --- space exploration --- send astronauts into space - for example: Mars ----- Spending a long time in the zero-gravity environment has negative effects on the human body, such as decreased muscle mass and lower bone density ---- bones become weaker and more likely to break -- many elderly people also have weak bones -- develop drugs -- first be tested on astronauts in space -- effective -- senior citizens --- improve life on Earth.

地球有一天会毁灭 Earth will become too crowded and too hot -- not suitable for human -- need to live on Mars someday --- it is necessary to spend money on space program

我们的地球在未来的某一时刻可能会因为核战争 nuclear war 和小行星 asteroid crash 撞击而毁灭 destroyed\ 也可以说太热太挤-Earth will become too crowded and too hot---- 为世界末日 doomsday 做准备 prepare for 是明智的 advisable，那么寻找地球的替代品 substitute \alternative planets 就是第一步。---比如，有一些国家在评估移居去火星 Mars 的可能性 或者 need to live on Mars someday：，他们相信在那里可以找到水源来维持人类生存 to ensure the survival of human beings----艺术品 artifacts，文物 culturally important objects，庄稼 crops 还有动物 animals 都可以送去这个新星球，以保证可以把它们留给后人 preserved for future generations。

b **Source:** A newspaper editorial

The US has spent more than $16 billion per year on its space programme since 1958. For a long time, the cost was worth it because of the advances made in science and technology. However, space travel is not only expensive, but it is also dangerous. Astronauts have been killed as recently as 2014. In fact, as of 2022, 19 astronauts have died in space and 4 more have died before flight. Also, we shouldn't waste natural resources on building new spacecraft. More money should be spent on people who need clean water and food, on access to education and on medical research. For example, a Japanese study found that a drug made from sea sponges helps treat several types of cancer. And yet 90% of the Earth's oceans are still unexplored. The ocean is a valuable resource. Governments should spend more money on ocean exploration than on space exploration.

不支持外太空探索

危险性太强了 ： space travel is dangerous -- astronauts have been killed --- 大肆来讲： 外太空对人的危害--- astronauts on a mission to Mars would be exposed to dangerous levels of space

radiation, much of which comes in the form of charged particles emitted by the Sun. 或者可以直接写：对身体的负面影响 Spending a long time in the zero-gravity environment has negative effects on the human body, such as decreased muscle mass and lower bone density

waste - save

space exploration is too expensive --- 浪费 waste natural resource on building spacecraft --- more money should be spent on people who need clean water and food\ on access to education\ on medical research --- fundamental social sectors --- 对比分析： ocean exploration: Earth oceans - sea sponge -- develop drugs --- help treat several types of cancer -- governments should spend more money on ocean exploration rather on space exploration

当你的思路是花钱、花时间、花空间
立即补充省下来时间、钱、空间干什么
必须有比较的思维， 你的论证才能合理

a

Source: The International Space Agency

Governments around the world should spend more money on space programmes. The International Space Station (ISS) is a good example why. The ISS has existed since 1998 and brings together many countries. Astronauts live in space and take part in important experiments. In 2022, astronauts from Japan and Canada went to the ISS for six months to observe the effect of space on the human body. Long-term journeys, such as travelling to Mars, will never happen without this important research. The ISS cost about $100 billion and one country could not pay for that on its own. International space exploration proves that countries can work together. It represents the spirit of partnership. With so much war in the world, governments should spend more money on things they can achieve together. Maybe that will help bring peace.

支持外太空探索
政府应该投资更多资金---比如国际空间站---汇集多个国家---如：日加共同研究太空对人体影响---如：国际站的耗资单一国家无法支撑---合作精神的标志---有利和平
cost: huge --- one country cannot pay for that on its own -- countries should work together to figure out the problem ---- represents the spirit of partnership -- cooperation --- reduce wars in the world -- bring peace

passage 4

IS EXPLORING SPACE A WASTE OF TIME AND MONEY?



NASA satellite image of Typhoon Goni

Countries around the **globe** are spending billions of dollars and lots of time on various space missions, whether to Mars or other planets much further away. Some people **argue** that we should stop wasting time and money exploring space. Instead, we should feed the world's poor and find immediate solutions to other problems, such as pollution and fatal diseases. However, others feel this is a **shallow** view which fails to realise how exploring space helps us.

Firstly, exploring space has already made a difference in the fight against world hunger. It has directly resulted in the many satellites that now orbit Earth. A number of the satellites record data on land and weather **patterns**. Then the data is transmitted to scientists on Earth. After careful analysis, the scientists can provide useful recommendations and advice for farmers. As a result, space-based science has helped farming in its efforts to grow enough food to feed Earth's increasing population.

Secondly, space exploration has already promoted technological improvements that benefit us all. High-end products around the world are made to a higher standard now because of advanced technology which was first created to meet the requirements for space exploration. For example, space technologies have helped the research and development of different types of new material. They have also helped companies make better heart **monitors** and other machines that doctors regularly use. Today, space technologies are widely used in all kinds of industries, and everyday products such as GPS, memory foam pillows, and smartphone cameras are changing our lives.

Finally, sending astronauts into space has helped people to think about the world's problems and even to find ways to solve them. Seeing pictures of our planet as an island in a black sea made people realise that our planet's **resources** are **limited**. In order to provide for such a rapidly increasing population, scientists are trying to find other planets that could one day be our new home. The greatest attention at present is on Mars because it is closer to Earth. In the future, humans may live on both planets.

In closing, exploring space provides the world with many different benefits. Therefore, it should continue so as to provide new and better solutions to people's short-term and long-term problems.

驳斥段的过渡句 however, it is a shallow view which fails to do sth.
However, it is not the case.
However, it is an incomplete part of the picture.
However, it ignores \ does not mention ....

驳斥段的思路:

反方的观点 some people argue that... + 理由 + 展开 + 怼回去 however, it is a shallow view which fails to do sth. + 进一步的展开

驳斥段的例子

反方观点 space exploration is too expensive --- 浪费 waste natural resource on building spacecraft --- 更多的钱用在基础的板块 more money should be spent on people who need clean water and food\ on access to education\ on medical research --- fundamental social sectors ---- however, it is a shallow view which fails to do sth. + 进一步的展开 fight against world hunger --- 具体的展开 many satellites orbit Earth --- record data on land and weather patterns --- the data is transmitted to scientists on Earth -- careful analysis -- scientists can provide useful recommendations and advice for farmers --- space - based science helps farming in its effort to grow enough food to feed Earth's increasing population

不进行外太空探索

waste time and money --- save the cost -- feed the world's poor and find immediate solutions to other problems---- pollution and fatal diseases

支持外太空探索

解决饥饿问题 space exploration can make a difference in the fight against world hunger --- many satellites orbit Earth --- record data on land and weather patterns --- the data is transmitted to scientists on Earth -- careful analysis -- scientists can provide useful recommendations and advice for farmers --- space - based science helps farming in its effort to grow enough food to feed Earth's increasing population

促进科技 space exploration -- technological improvements -- benefit humans -- high -end products : first meet the requirement for space exploration -- Earth: advanced technology ---- new material: for example: better heart monitor and other machines that doctors regularly use -- another example: in all kinds of industries -- every products-- GPS\ smartphone cameras

寻找到解决问题的方法： help people to think about the world's problems and eve to find ways to solve them -- planet's resources are limited -- cannot meet the basic demand of the increasing population -- find other planet that could be one day be our new home -- Mars

补充阅读：外太空旅游

SPACE TRAVEL

NEWS SCIENCE DISCOVERIES TECHNOLOGY EDUCATION BLOGS COMMUNITY



Space craft orbiting Earth

1 In 1957, the Soviet Union sent *Sputnik I* into space. It was the first successful spacecraft to orbit[2] Earth, and it started the time period known as the Space Age. A short time later, in the United States, the National Aeronautics and Space Administration (NASA) successfully sent another spacecraft, *Explorer I*, into space. In the years that followed, incredible **advances** were made. Astronauts orbited Earth and men walked on the moon. The world, it seemed, wanted to learn what was **beyond** Earth.

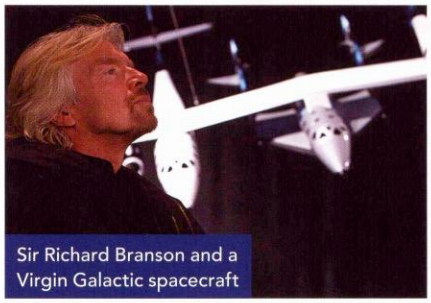
2 Today, space exploration continues, and governments still compete with one another to make new discoveries. In 2012, NASA landed its unmanned[3] spacecraft *Curiosity* on Mars in order to collect information about the planet. In 2016, Europe and Russia worked together and sent a spacecraft to Mars. China and India are also working on similar projects. However, there has been one big change: **private** companies, instead of **public** government organizations, are entering the Space Race[4].

3 SpaceX has been very successful in commercial space travel. They designed the spacecraft *Dragon* in order to deliver supplies to the International Space Station (ISS). In 2012, *Dragon* was the first commercial spacecraft in history to do that. Elon Musk, the man who started the company, has dreams that go beyond making deliveries. He hopes to build a base on Mars by 2028 and a one million-person city there in the next 50 to 100 years.

4 Another **entrepreneur** who supports commercial space travel is Sir Richard Branson. He started a private company called Virgin Galactic. Their goal is to open space travel to everyone. The company has sold about 700 future trips to space, some at the high cost of $450,000 per person. Those future space tourists come from countries all over the world and are all different ages.

5 Private companies are lucky in one way. They don't have to wait for money from the government like NASA does. However, that doesn't mean that setbacks[5] and accidents don't happen. In 2014, Virgin Galactic's *VSS Enterprise* **crashed** in the Mojave Desert in the US during a test flight. The 39-year-old pilot, Michael Alsbury, was killed. In 2023, the SpaceX spacecraft *Starship* exploded on its first flight after around two or three minutes. No one was hurt and Elon Musk said they were going to try again in a few months.

6 The race to **explore** the universe continues and many private companies are competing. Some of those companies want to take people to the moon and back someday. Others want to take people to Mars. The possibilities are endless. Maybe in our lifetime, those dreams will come true.



Sir Richard Branson and a Virgin Galactic spacecraft

作业 1:上课笔记

作业 2: 整理所有的逻辑链条 - 同意有哪些， 不同意有哪些?

x- mind

作业 3:任选一题，必须用到我的素材

word 文档 - 标记的你的名字 海燕- 外太空

Some people think that money spent on developing technology on space exploration is not justified. They think there are more beneficial ways to spend money. Do you agree or disagree?

Some people think the government should spend money to look for life on other planets, while others think it is a waste of public money, because there are other problems unsolved on Earth. Discuss both sides and give your opinion.

Some scientists believe intelligent life forms may exist in other planets and some want to send messages to the space to contact with these life forms. But other scientists believe it is a bad idea and it could be dangerous. Discuss both ideas and give your own opinion.

作业 4: 学有余力的小伙伴 -- 补充材料

整理生词， 概括段落大意， 做成思维导图